

Experiment - 06

Student Name: Bharti Mehra
Branch: CSE - AIML
Semester: 5
Subject Name: ADBMS

UID: 24BAI70051
Section/Group: 24AIT_KRG-1_B
Date of Performance: 10/8/26
Subject Code: 24CSP-310

(A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

Solution:

```
select sale_date,  
sum(  
case when sales.fruit='apples'  
then sold_num
```

```

else -sold_num
end
) as diff
from sales
group by sale_date
order by sale_date;

```

(B)

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

id represents the current node.

p_id represents the parent node.

Classify every node into one of the following categories:

Root: A node that does not have any parent.

Leaf: A node that has a parent but does not have any children.

Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Tree	
id	p_id
1	NULL
2	1
3	1
4	2
5	2

Expected Output

id	Type
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf

Solution:

```

select id,
case when tree.p_id is null then 'Root'
when id not in (select p_id from tree where p_id is not null) then 'Leaf'
else 'Inner'
end as Type
from tree;

```